

Name: \_\_\_\_\_

## Practice Quiz for CS599 B1: Advanced Natural Language Processing

### Instructions:

- You will have 20 minutes at the start of class to complete the quiz.
- This quiz is to be completed individually by each student.
- You are **not** allowed any reference material, calculators, or other electronic devices.
- Answer these questions by circling the correct option or writing the answer. **If given letter options, give exactly one answer. Carefully read the instructions on each question.**

1. (10 points)

\_\_\_\_\_ **D** 1. What is the correct order of operations in the Transformer block you implemented? (Note: this is *not* the one you implemented in your ablation experiments.)

- A. RMSNorm, SwiGLU MLP, RMSNorm, Self-attention
- B. Self-attention, RMSNorm, SwiGLU MLP, RMSNorm
- C. SwiGLU MLP, RMSNorm, Self-attention, RMSNorm
- D. RMSNorm, Self-attention, RMSNorm, SwiGLU MLP

\_\_\_\_\_ 2. Let  $d_k$ ,  $d_v$ , and  $d_{\text{model}}$  denote vector dimensions per head, and let  $h$  denote the number of heads. How many parameters are in the multi-head self attention layer? Do not make any assumptions about the relation between  $d_k$  and  $d_{\text{model}}$ . Ignore bias terms in linear layers.

*$2d_{\text{model}}hd_k + 2d_{\text{model}}hd_v$ . The first term is from the  $Q$  and  $K$  matrices (hence the 2). The second term is from the  $V$  and  $O$  matrices.*

\_\_\_\_\_ **B** 3. Which of the following is the most complete and accurate description of what a SwiGLU contains? Select only one.

- A. A sigmoid activation function
- B. A Swish activation function and linear gate
- C. A linear gate
- D. A Swish activation function, linear gate, and ReLU activation function

\_\_\_\_\_ 4. Given the string “l a t e \_ l a t s \_ l a m p” that has been split into characters, what would be the first **two** tokens added to the vocabulary when running BPE? Assume that you cannot merge “\_” characters, nor merge across them.

*The first merge is (“l”, “a”). The second is (“la”, “t”).*

\_\_\_\_\_ **A** 5. When using rotary position embeddings (RoPE), which of the following is true?

- A. The key at position 5 and query at position 10 will have the same positional relationship as the key at position 15 and query at position 20.
- B. The token at position 100 will have the same RoPE encoding as the token at position 1000.
- C. RoPE is added to the input embeddings before the first Transformer block.
- D. At a given position, every dimension in the query vector is rotated to the same degree.

\_\_\_\_\_ 6. Given the vocabulary [a, b, c, d, e] with probability distribution [0.1, 0.4, 0.3, 0.05, 0.15], which tokens could you sample if you were to use nucleus sampling with  $p = 0.9$ ?

a, b, c, e.

\_\_\_\_\_ 7. In 1–2 sentences, give one concrete reason why we might want to train a BPE tokenizer on UTF-8 bytes instead of characters.

There are a few possible reasons. UTF-8 is guaranteed to not yield out-of-vocabulary tokens at reasonable vocabulary sizes. Relatedly, UTF-8 is better for multilingual models, especially for languages like (for example) Japanese, which have many characters.

\_\_\_\_\_ C 8. As a function of the number of parameters  $P$  in your language model, how does the memory usage of AdamW scale?

- A.  $O(1)$
- B.  $O(\sqrt{P})$
- C.  $O(P)$
- D.  $O(P^2)$

\_\_\_\_\_ C 9. For which of the following operations is it most important to use `float32` precision, rather than `float16`?

- A. Linear layer
- B. The Swish function
- C. RMSNorm
- D. The input embedding layer

\_\_\_\_\_ B 10. What is the correct order of operations in AdamW?

- A. Compute gradient, compute and apply weight update, perform moment updates
- B. Compute gradient, perform moment updates, compute and apply weight update
- C. Perform moment updates, compute gradient, compute and apply weight update
- D. Perform moment updates, compute and apply weight update, compute gradient